Total no. of Pages: 2 Roll no. .....

Degree: B.Tech. Semester: 3 Course work

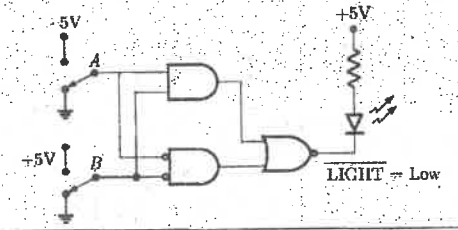
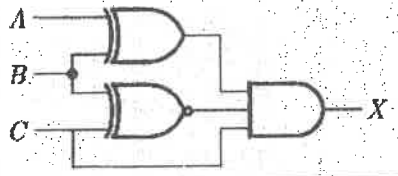
MID-SEMESTER EXAMINATION, SEP 2026

Course Title: Digital Circuits and Systems

Course Code: ECECC305

Duration: 1:30 Hours Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

	Questions	Marks	CO
1a	Consider the given circuit diagram showing the switching of an LED using two switches. Determine the input combinations of the switches required to turn ON the LED. 	2	CO1
1b	In circuit shown below, for what input at the terminals A, B and C the output is X= 1? 	2	CO1
2a	State any two advantages of the 2's complement method over the 1's complement method. Perform the arithmetic operation $(12-35)_{10}$ using the 2's complement method.	2	CO1
2b	Prove the following Boolean algebra identity using Boolean theorems and postulates: AB+A'C+BC=AB+A'C	2	CO1
3a	Using the tabulation (Quine–McCluskey) method, simplify the following Boolean function: F(w, x, y, z)=Σm(0,3,5,6,7,10,12,13)+Σd(2,9,15)	2	CO1
3b	Construct a 4-bit BCD adder using two 4-bit parallel adders and appropriate logic gates. Show the correction logic and explain its operation.	2	CO2
4a	Construct a 5-to-32-line decoder using four 3-to-8-line decoders with enable inputs and one 2-to-4-line decoder. Use block diagrams for the components and show all necessary connections. Clearly indicate the inputs and outputs.	2	CO2
4b	Design a combinational circuit that accepts a 3-bit number as input and generates an output binary number equals to square of input number.	2	CO2
5a	A switching network has four inputs A, B, C, and D. A and B represent the first and second bits of the binary number N_1, respectively, while C and D represent the first and second bits of the binary number N_2, respectively. The output of the network is 1 only if the product $N_1 \times N_2$ is less than or equal to 2. Implement the network using a 4 × 1 multiplexer and external gates.	2	CO2
5b	Implement the following two Boolean functions with a PLA: $F_1(A, B, C) = \sum m(0,1,2,4)$ and $F_2(A, B, C) = \sum m(0,5,6,7)$	2	CO2